

$$= \begin{bmatrix} 8+(-5) \\ 12+(-1) \end{bmatrix} \text{ (Vector addition)} = \begin{bmatrix} 3 \\ 11 \end{bmatrix}$$

$$Av = \begin{bmatrix} 2 & 5 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} -3 \\ 5 \end{bmatrix} = -3 \begin{bmatrix} 2 \\ 3 \end{bmatrix} + 5 \begin{bmatrix} 5 \\ 1 \end{bmatrix} \text{ (Defn. of } Ax)$$

$$= \begin{bmatrix} -3 \cdot 2 \\ -3 \cdot 3 \end{bmatrix} + \begin{bmatrix} 5 \cdot 5 \\ 5 \cdot 1 \end{bmatrix} \text{ (Scalar multiplication)}$$

$$= \begin{bmatrix} -6 \\ -9 \end{bmatrix} + \begin{bmatrix} 25 \\ 5 \end{bmatrix} = \begin{bmatrix} -6+25 \\ -9+5 \end{bmatrix} \text{ (Vector addition)} = \begin{bmatrix} 19 \\ -4 \end{bmatrix}$$

$$A(u+v) = \begin{bmatrix} 3 \\ 11 \end{bmatrix} + \begin{bmatrix} 19 \\ -4 \end{bmatrix} = \begin{bmatrix} 3+19 \\ 11+(-4) \end{bmatrix} \text{ (Vector addition)} = \begin{bmatrix} 22 \\ 7 \end{bmatrix}$$

$$\therefore A(u+v) = Au + Av. \text{ We have verified Theorem 5(a).}$$

3. Construct a 3×3 matrix A and vectors b and c in $\mathbb{R}^3$ so that $Ax=b$ has a solution, but $Ax=c$ does not.

Solution:

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 4 & 9 & 7 \\ 5 & 11 & 6 \end{bmatrix}$$

$$b = \begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix}$$

$$c = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$$

Claim: For the given A and b , $Ax=b$ has a solution.

Proof: By Theorem 3, the solution of $Ax=b$ is the same as the system of equations corresponding to the augmented matrix.

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 2 \\ 4 & 9 & 7 & 3 \\ 5 & 11 & 6 & 5 \end{array} \right]$$

$$\begin{array}{l} \text{Row 2} = \text{Row 2} - 4 \times \text{Row 1} \\ \text{Row 3} = \text{Row 3} - 5 \times \text{Row 1} \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 2 \\ 0 & 1 & 11 & -5 \\ 0 & 1 & 11 & -5 \end{array} \right]$$

$$\begin{array}{l} \text{Row 3} = \text{Row 3} - \text{Row 2} \\ \text{Row 3} = \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 2 \\ 0 & 1 & 11 & -5 \\ 0 & 0 & 0 & 0 \end{array} \right]$$